RETAIL SALES ANALYTICS

SQL Analysis & Power BI Reporting Project

Final Project Report

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Tools: SQL • Power BI • DAX • Excel

Reporting period: January 2023 – December 2025

Portfolio project using a synthetic retail dataset. The results demonstrate an analytical workflow and are not claims about a real company.

1. Executive Summary

This project analyzes 1,800 retail order records across a three-year period to demonstrate how SQL and Power BI can support commercial decision-making. The dataset contains 250 distinct customers and 5,482 units sold. Total recorded sales are **$844,294.53**, total profit is **$210,259.85**, and overall profit margin is **24.90%**. The work covers performance KPIs, time trends, category and regional comparisons, discount analysis, customer ranking, and basic data-quality checks.

Total Sales	Total Profit	Profit Margin
$844,295	$210,260	24.90%
Orders	**Customers**	**Units Sold**
1,800	250	5,482

2. Business Problem and Objectives

Business problem: retail stakeholders need a consolidated view of revenue, profitability, product performance, customer contribution and discount patterns. Looking at sales alone can hide low-margin activity, so the analysis pairs revenue with profit and margin.

• Measure overall performance using sales, profit, margin, orders, customers and units sold.

• Compare performance by month, product category, subcategory, region, customer and shipping method.

• Investigate whether discount bands coincide with changes in profitability.

• Use SQL to make results reproducible and Power BI to make them interactive.

• Identify data-quality checks that should be performed before business reporting.

3. Dataset and Methodology

The source file is **retail_sales.csv**, containing 1,800 rows and 14 columns. Fields include order ID/date, customer ID, segment, region, state, category, subcategory, product ID, quantity, discount, sales, profit and shipping mode. Dates span 2023-01-03 to 2025-12-30.

Data provenance: the dataset was generated synthetically for this project. It is not downloaded from a company or public data portal. The findings are therefore suitable for demonstrating methods, not for making claims about actual retail market performance.

Workflow: inspect and load the CSV; aggregate and rank records with SQL; calculate KPIs; validate data quality; create a Calendar table and DAX measures in Power BI; then visualize trends and comparisons.

4. SQL Analysis

The companion file **analysis_queries.sql** includes ten query examples. These cover:

KPI summary: Distinct orders/customers, total sales/profit, margin and units.

Monthly trends: Sales and profit grouped by year-month.

Category and region: Sales/profit comparisons and ranking.

Subcategory ranking: Highest-sales subcategories.

Discount bands: Order-line counts, sales, profit and margin by discount range.

Customer ranking: Top customers and dense ranking by sales.

Running total: Cumulative monthly sales using a window function.

Data quality: Blank IDs, negative sales/quantity and invalid discount values.

Shipping performance: Sales and profit by shipping method.

Example KPI query

```
SELECT COUNT(DISTINCT OrderID) AS orders,
SUM(Sales) AS total_sales,
SUM(Profit) AS total_profit,
100.0 * SUM(Profit) / NULLIF(SUM(Sales), 0) AS margin_pct
FROM retail_sales;
```

5. Results and Findings

Category performance: Technology is the largest category by sales at $433,487.76.

Regional performance: West leads the regions by sales at $246,536.06. Regional profit should also be reviewed before allocating more resources.

Subcategory performance: Accessories is the leading subcategory by sales at $123,477.94.

Discount analysis: the Above 20% discount band has the lowest observed margin in this generated dataset (4.48%). This is an association in simulated data, not evidence that discounts alone caused lower profitability.

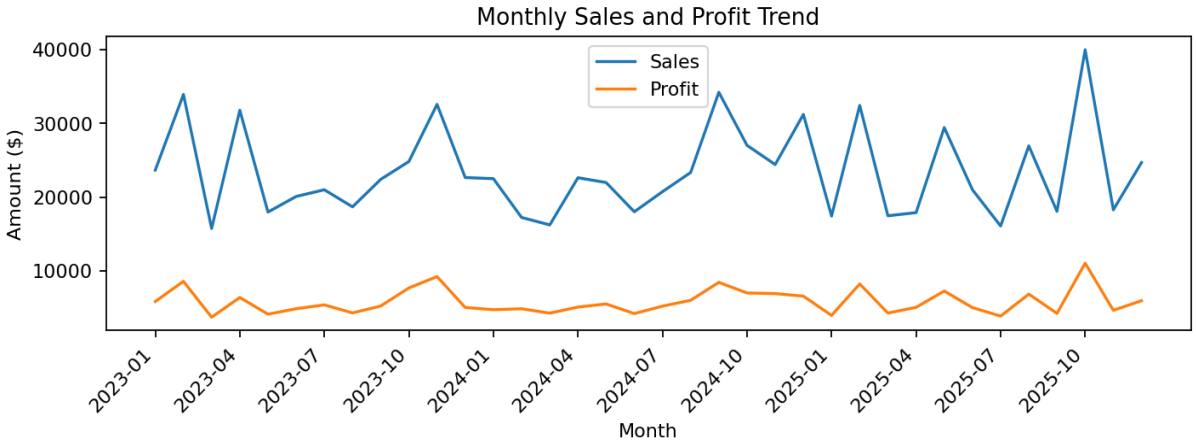


Figure 1. Monthly sales and profit trend from the synthetic dataset.

6. Visual Analysis

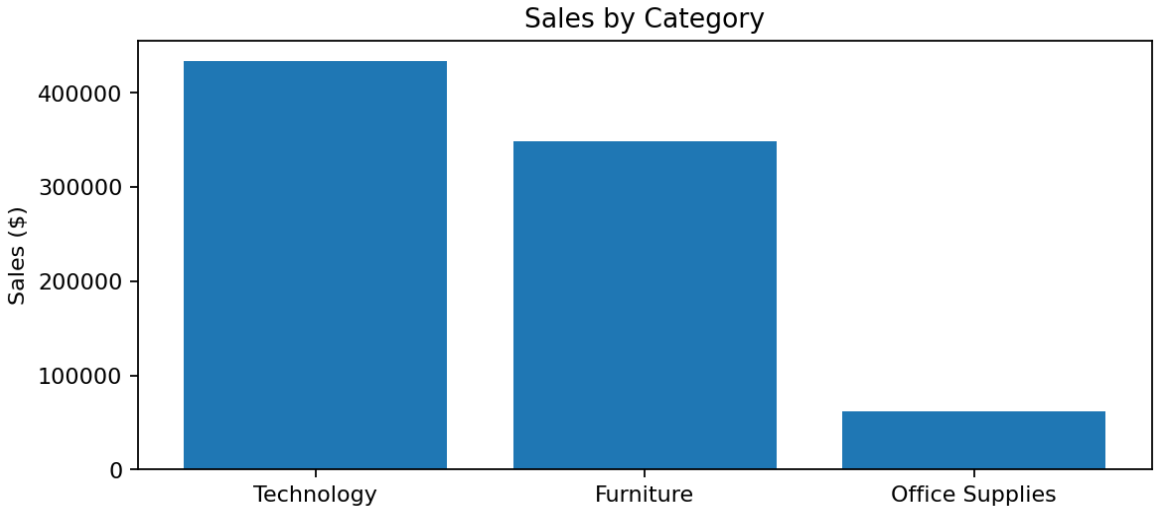


Figure 2. Total sales by category.

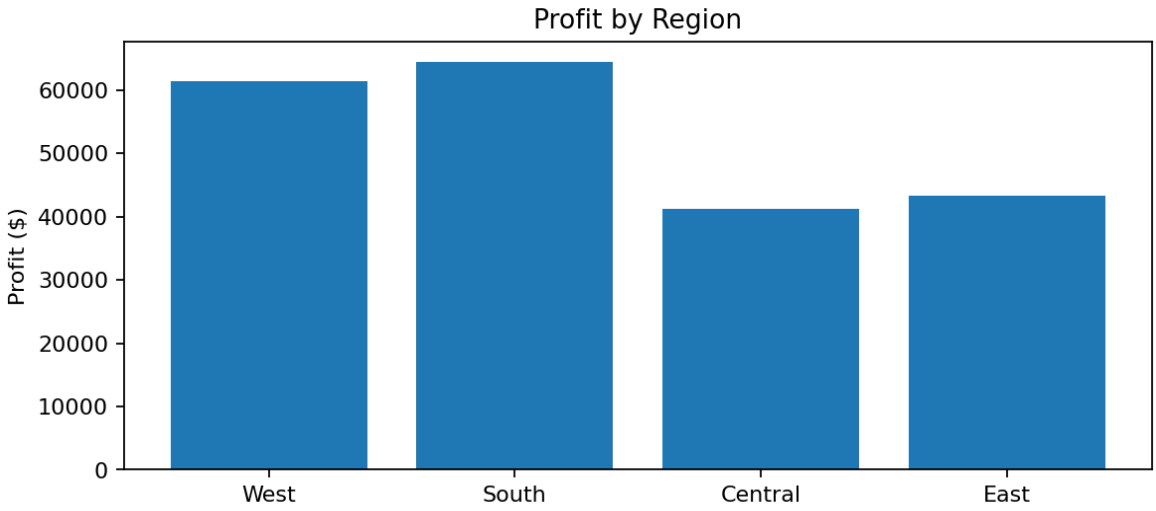


Figure 3. Total profit by region.

7. Power BI Report Design

The intended Power BI report contains three pages. The PDF charts are static analysis visuals; the interactive report should be completed and saved separately as a .pbix file in Power BI Desktop.

Report page	Recommended visuals	Business purpose
Executive Overview	KPI cards; monthly sales/profit line chart; sales by category; profit by region; slicers	Understand overall performance and trends.
Product & Discount	Category/subcategory matrix; top 10 subcategories; discount vs profit scatter; margin by discount band	Find product opportunities and profitability risks.
Customer & Operations	Top-customer table; shipping performance; optional state map; segment contribution	Understand customer and operational patterns.

Core DAX measures

• Total Sales = SUM(retail_sales[Sales])

• Total Profit = SUM(retail_sales[Profit])

• Profit Margin % = DIVIDE([Total Profit], [Total Sales], 0)

• Total Orders = DISTINCTCOUNT(retail_sales[OrderID])

• Average Order Value = DIVIDE([Total Sales], [Total Orders], 0)

8. Recommendations

• Monitor profit and margin alongside sales; revenue growth does not always mean stronger profitability.

• Review category and subcategory combinations with weaker margins before expanding promotions.

• Use monthly trend analysis to identify periods where sales rise but profit does not rise at the same pace.

• Compare regions using both sales and profit to avoid optimizing for revenue alone.

• Refresh the data-quality checks whenever new data is added, and reconcile Power BI totals with SQL outputs.

9. Data Quality and Limitations

Basic checks found 0 blank order IDs, 0 negative-sales rows, 0 negative-quantity rows and 0 discounts outside the 0–1 range. These checks are limited and do not establish that every field is correct. The dataset is synthetic, the financial amounts are illustrative, and any observed relationships are not causal conclusions. The Power BI report must be built and validated in Power BI Desktop before the project is described as fully interactive.

10. Conclusion

This project demonstrates an end-to-end analytics workflow: structuring retail data, answering business questions with SQL, calculating measures for reporting, and translating results into dashboard recommendations. The next implementation step is to finish the interactive Power BI pages, validate all metrics against the SQL outputs, and include screenshots of the completed report in the portfolio.

Appendix: companion files

• retail_sales.csv — source dataset

• analysis_queries.sql — SQL analysis queries

• schema_and_setup.sql — table schema

• powerbi_dax_measures.txt — DAX measures

• powerbi_build_guide.md — dashboard construction guide

• sql_results.xlsx — reference query results

• README.md — project overview